



Vel Tech
Rangarajan Dr. Sagunthala
R&D Institute of Science and Technology
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Avadi, Chennai

Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology
School of Computing

Department of Computer Science and Engineering

10211CS224 / FULL STACK APPLICATION DEVELOPMENT (JAVA)

END SEMESTER REVIEW

Date: 06-05-2026

Smart Campus Event Management System

SDG: SDG No – SDG Title(Example: SDG 4 – Quality Education)

Presented By:

CH.Lavanya – VTU26565– Reg.No.23UECS0120

Courses Handling Faculty:

Dr.Hafizur Rahman

Assistant Professor

INTRODUCTION



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- In modern campuses, managing events efficiently is a major challenge.
- Students often miss important updates due to scattered communication.
- This system provides a **smart centralized platform** for managing all campus events.
- It allows students to explore, register, and track events easily.
Admins and organizers can manage events with automation and real-time updates.
- The system improves communication and engagement across the campus.
- It ensures all event-related information is accessible anytime.
This solution makes campus event handling more efficient and organized.

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PROBLEM STATEMENT



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- Event information in campuses is often unorganized and delayed.
- Students depend on notice boards, WhatsApp groups, or emails.
- There is no unified platform for event registration and tracking.
- Manual handling causes confusion and missed opportunities.
- Organizers face difficulty in managing participants and updates.
- Lack of real-time updates reduces participation.
- A smart digital system is needed to solve these issues efficiently.

OBJECTIVE(S)



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- To develop a smart web-based campus event management system
- To provide real-time updates and notifications
- To simplify event registration and participation
- To improve communication between students and organizers
- To create a user-friendly and responsive interface

PROPOSED SYSTEM



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- A centralized web-based application to manage all campus events efficiently in one platform

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- Built using Spring Boot architecture to ensure scalability, performance, and maintainability

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- Secure login system with role-based access for Admin and Student users
- Dynamic event management including creation, updates, and real-time event display
- Admin can add, edit, and delete events while students can view and register for events
- System securely stores event and registration data with proper backend processing

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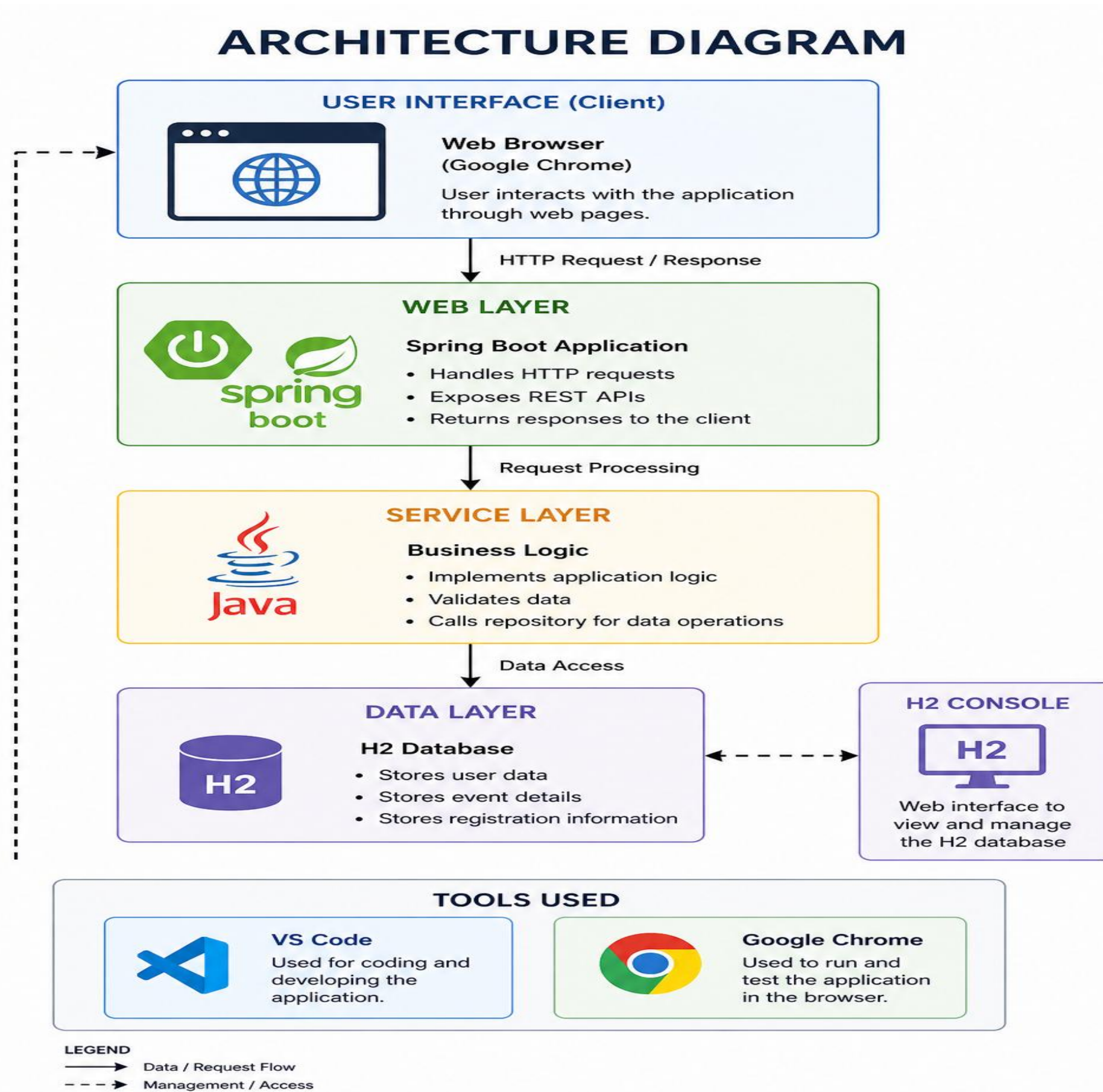
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PROPOSED SYSTEM ARCHITECTURE

- The system follows a **three-tier architecture** consisting of Presentation Layer, Application Layer, and Database Layer
- It ensures smooth communication between user interface, backend logic, and data storage
- Provides user interface for Admin and Students using HTML, CSS, and Thymeleaf
- Displays event details, registration forms, and dashboards dynamically
- Developed using Spring Boot to handle business logic and request processing
- Uses Controllers and Services to manage event operations and user interactions
- Implements authentication and authorization using Spring Security
- Ensures secure login with role-based access control (Admin & Student)
- Uses MySQL / H2 database to store event, user, and registration data

Architecture Diagram



WORKFLOW



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- User accesses the system through a web browser and logs in using valid credentials
- The system authenticates the user and grants access based on role (Admin or Student)
- Admin creates, updates, or deletes event details through the dashboard
- Event data is stored and managed in the database
- Students browse available events and view detailed information
- Students register for selected events through the system
- The system processes requests and displays updated event and registration information in real time

TECHNOLOGIES USED



Frontend Technologies

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HTML – Used to design the structure of web pages such as login, dashboard, and event pages

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CSS – Used for styling, layout design, and improving user interface appearance

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JavaScript - Adds interactivity and dynamic behavior.

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Backend Technologies

Spring Boot – Main framework used to build the application and handle backend development

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Java – Used as the core programming language to build backend logic

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Database

H2 Database – Used to store user data, event details, and registration information

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H2 Console – Used as a web interface to view and manage the H2 database

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Tools & Development Environment

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VS Code – Used for coding and developing the application

Google Chrome – Used to run and test the application in the browser

MODULE DESCRIPTION



User Authentication Module

Provides secure login functionality for both Admin and Student users

Implements role-based access control using Spring Security

Validates user credentials and prevents unauthorized access

Ensures secure session handling and authentication process

Event Management Module

Allows Admin to create new events with complete details (title, date, description)

Enables updating and modifying existing event information

Supports deletion of events that are canceled or outdated

Maintains all event data in the database for easy access and management

Event Registration Module

Allows students to register for available events through a simple interface

Stores participant details securely in the database

Helps organizers track the number of registrations

Simplifies the registration process compared to manual methods

Backend Processing Module

Handles all client requests using Spring Boot controllers

Processes business logic through service layer components

Validates input data before storing or processing

Ensures smooth communication between frontend and database

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IMPLEMENTATION SCREENSHOTS



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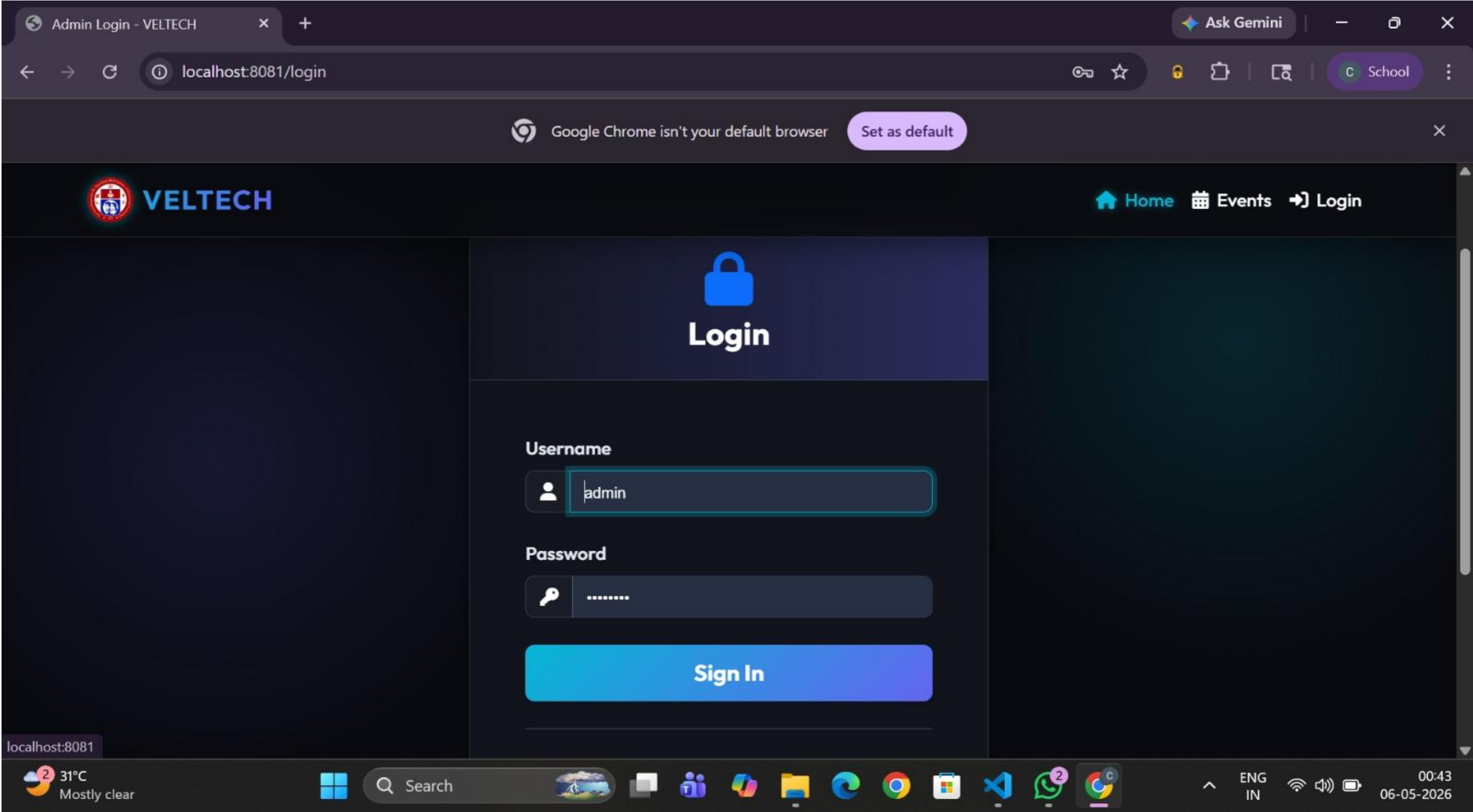
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IMPLEMENTATION SCREENSHOTS



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Create Account - VELTECH

localhost:8081/signup

Google Chrome isn't your default browser Set as default

VELTECH

Home Events Login

Create Account

Full Name

Your full name

Email Address

your.email@college.edu

Username

@ admin

Must be 3-20 characters. Letters, numbers, and underscores only.

Password

31°C Mostly clear

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IMPLEMENTATION SCREENSHOTS



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Admin Dashboard - VELTECH

Ask Gemini

localhost:8081/admin

Google Chrome isn't your default browser

Set as default

HomeEventsDashboardLogout

+ Add New Event

Event Name

Department

Type

Conference

Date

dd-mm-yyyy

Seats

0

Venue

Manage Events

Name	Dept / Type	Date	Seats	Actions
Tech Symposium 2026	Computer Science Conference	2026-06-15	500	
Annual Sports Meet	Physical Education Sports	2026-07-20	1000	
Robotics Workshop	Electrical Eng Workshop	2026-05-10	50	
Cultural Fest - Utsav	Arts Festival	2026-08-05	2000	
AI/ML Bootcamp	Computer Science Workshop	2026-06-01	40	

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ADVANTAGES & LIMITATIONS



ADVANTAGES

- Provides a **centralized platform** to manage all campus events in one place
- Reduces dependency on manual methods like notice boards and messages
- Ensures **real-time access** to event updates and information
- Improves communication between students and event organizers
- Simplifies event registration process through an online system
- Enhances student participation by making events easily accessible

LIMITATIONS

- Requires **stable internet connection** to access the system
- Basic user interface may not be highly interactive or modern
- Limited to web platform (no mobile application support)
- Notification system (email/SMS) not fully implemented

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Thank you

